

DishKit - Integrated Dish Preparation and Ingredient Delivery System

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Abstract. Long-term productivity and personal health depend on having a healthy diet. However, this is jeopardised in the modern world due to time consuming processes like gathering raw materials according to cooking size, cooking time, and not enough knowledge of cooking instructions. Thus, there is a greater propensity to purchase fast food, order food online, and eat out. In this paper we propose an integrated dish preparation and ingredient delivery system. By cutting down on raw material collection time and offering cooking instructions, the method helps encourage people to cook at home. In addition to making grocery shopping simpler, this initiative aims to address the growing need for healthier eating habits by facilitating the preparation of wholesome, fresh meals at home. The approach is implemented as a web application, DishKit. Preliminary findings indicate a significant rise in the number of consumers opting for home cooking that prioritizes convenience and wellness.

Keywords: Recommendation system, Feedback analysis, Application Program interface, Nutrients calculation, Natural Language Processing(NLP), Healthy food

1 Introduction

In today's busy world, people often choose what's convenient to eat over what's nutritious, common trends in food ordering reveal a growing reliance on outside sources for meals [9]. Busy lifestyles often lead to people choosing fast food or restaurant meals over cooking at home, mainly due to the time commitment required for shopping and cooking. Many are also discouraged by not knowing the right amount of ingredients or cooking methods. Meal kit delivery services address these issues by sending pre-measured ingredients and recipes directly to subscribers, making home cooking more convenient and varied without the need for meal planning or grocery shopping. These services have become popular due to their ease and the chance to experiment with new recipes. In this paper, we are proposing a variant of meal kit service, *an integrated dish preparation and ingredient delivery system, DishKit*. Reflecting on this, a study [4] on the factors associated with ordering food via online meal ordering services identifies younger individuals, those with higher education and income, and frequent consumption of sugary drinks and fast food as key users, highlighting a pivotal shift in meal acquisition habits

A survey studied the effects of eating habits on health, weight, and finances, examining how often people eat out versus at home. Utilizing the findings, we developed DishKit, a web framework that uses technology like Long short-term memory (LSTM) for predictive modeling, Machine Learning (ML) for nutrient calculation, and sentiment analysis to promote healthier eating and streamline meal preparation. Initial tests show DishKit effectively delivers its intended features.

In the next section, we outline the survey and the observations. The proposed approach is explained in Sec. 3. In Sec. 4 we present the implementation details. Our experimental evaluation is detailed in Sec. 5, followed by a discussion on related work in Sec. 7.

2 Survey

To study eating habits and their effect on individuals we have performed a survey on a set of individuals whose age is greater than 18 years. The data was collected via responses to a Google Form survey. The survey

participants include college students, lecturers, and IT professionals. A list of potential questions was prepared based on literature review. From this list, 15 questions were selected and modified for the questionnaire. The questions covered the following topics: frequency of eating outside food versus home-cooked meals, cooking interest, difficulties faced in home cooking, the financial impact of eating habits, weight gain, and health issues related to eating habits. Interesting observations from the survey are given in fig 1. On comparing

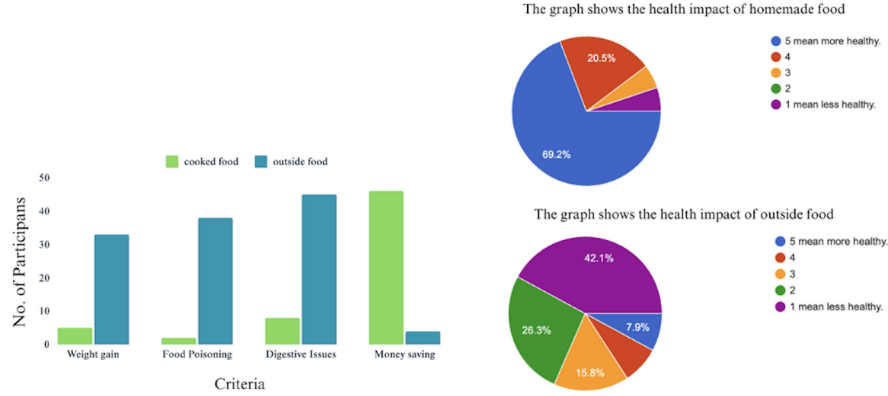


Fig. 1. Survey Result: comparing how consumers are affected by cooked and outside food

cooked food and outside food, it is observed that problems like weight gain, food poisoning, and digestive issues are more common for those who prefer outside food. When it comes to monthly expenses, cooked food is more money-saving. On consuming outside food, there is an increase in monthly expenses. The results are depicted in Fig 1. The healthImpact shows the health impact of homemade food and outside food. It is observed that homemade food is a healthier option compared to outside food. Other observations from the survey are: that the participants are interested in cooking, and if the time-consuming process of raw material gathering is reduced more individuals are interested in cooking. Another factor that restricts the participants in cooking is the lack of knowledge of cooking instructions and cooking time based on serving size. Also, most of the participants are well aware that eating from outside is not a healthier option.

The survey showed that young adults frequently dine out, which tends to affect both their weight and budget. Despite wanting to cook, they often lack the know-how. There's a notable enthusiasm for a service such as "*DishKit*," which could help by providing recipes and ingredients, encouraging more home cooking and less spending on eating out.

3 Proposed Approach

We provide an overview of the proposed framework and is outlined in Fig. 2. Our framework is developed to promote self-cooking behavior in society. Self-cooking aids in improving the long-term health issues and productivity of a person. It also helps in reducing weight gain and food-related short-term health issues. The user interacts with the framework to place an order and receive the ordered *DishKit*. DishKit is a package containing all the ingredients for a particular dish based on the number of people. The user can give the dish name. Then the next word predictor, predicts all possible dishes starting with the first word of the given dish name. In the next step, the ingredients for the selected dish are calculated based on the number of people. For each of the ingredients, nutritional value is calculated and displayed to the user. Finally, instructions and videos for cooking aid are selected based on the user and selected dish. In the next step, the nearest warehouse is selected based on the location. Similarly nearest delivery assistant is selected based on location. The delivery assistant can proceed with DishKit delivery to user. After delivery user can give feedback to the system. The feedback undergoes feedback analysis and is used to update the framework.

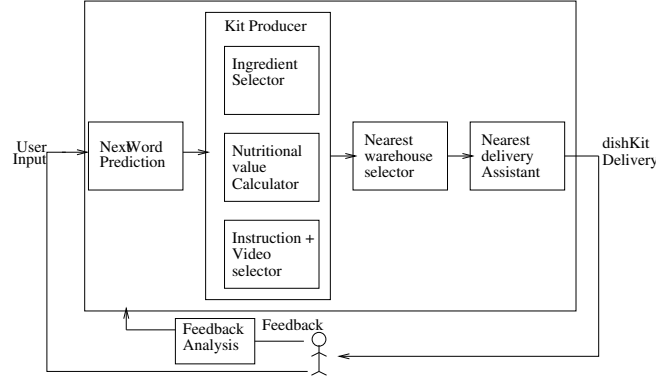


Fig. 2. Proposed Framework

The framework will have a user-facing interface and maintainer backend. The user side facilitates browsing, selection, and ordering of dish kits. The maintainer will update the framework based on the feedback from the users.

3.1 Next Word Prediction

The main objective here is to predict the next word of the dish name. In next-word prediction, we utilized predictive text modeling using Long Short-Term Memory (LSTM). The most likely words to appear in the position are listed by the model. The model predicts the next word as the user types the dish name, thereby accelerating the dish selection process.

3.2 Kit Producer

In this step, based on the selected dish and the serving size, the ingredients and quantity for each ingredient are selected from the Database. Using the ingredient quantities, the nutrient calculation model will estimate the nutrient content of the dish. The objective of this step is to inform users about the nutritional value of their chosen recipe. This will aid in improving the health-conscious decision-making of users. We employed a supervised learning model. The model uses Gradient Boosting Machines to predict the total amount of calories, proteins, fats, and carbohydrates present in a particular dish. The nutritional details are displayed as a pie chart. This will enhance the user's ability to comprehend nutritional data at a glance. In addition to the nutritional values, the cooking instruction for the particular dish is made available to the user which is already pre-entered into a Database. All the ingredient details for preparing the dish kit will be available after this step.

3.3 Nearest Location Calculator

The nearest location calculator is utilized in both selecting the nearest warehouse and finding the closest delivery assistant. It operates in such a way that the algorithm determines the nearest warehouse to the user based on the geolocation, that is, longitude and latitude. The distance is calculated using the Haversine formula [10].

After getting all the details regarding the ingredients from the previous step, the nearest warehouse is selected using the approach described before. The ingredients can be collected from the warehouse and delivered to the user by the nearest delivery assistant available.

3.4 Feedback Analysis

Once the dishKit is delivered, the user can provide their feedback. Sentiment analysis is used to interpret the emotional content of this feedback. It is integrated into the application's feedback loop, where user comments are processed, and sentiments are quantified. This data informs the recommendation engine and influences future updates to the framework.

4 Implemetation

4.1 Predictive Text Modeling Using LSTM:

On the DishKit platform, we’ve streamlined the dish selection process by enhancing the user interface with real-time predictive text suggestions. This is achieved through the implementation of Long Short-Term Memory (LSTM) networks, a sophisticated type of Recurrent Neural Network (RNN) that excels in processing and remembering sequential data over long periods. Our approach begins with the preprocessing of a dataset containing culinary terms and phrases, which involves tokenization, normalization, and vectorization to prepare the text for training. An Embedding layer is used to reduce dimensionality and provide meaningful word representations. The model architecture comprises an Embedding layer followed by Bidirectional LSTM layers, which capture contextual information from both directions, and concludes with a Dense output layer equipped with softmax activation for accurate word prediction. The training process is optimized with categorical cross-entropy to enhance the model’s predictive accuracy. Finally, through API-driven integration, the LSTM model interacts seamlessly with user inputs on the DishKit platform, offering predictive suggestions in real-time, thereby significantly improving the user experience.

4.2 Nutrient Calculation ML Model:

Our Nutrient Calculation ML Model on the DishKit platform is built on an extensive dataset of ingredients and their nutritional values to enable detailed nutritional analysis. We use Python’s Pandas for data cleaning and NumPy for numerical operations to preprocess the data, including normalizing ingredient quantities and using one-hot encoding for categorical data. The model is powered by Scikit-learn, utilizing its robust regression algorithms like Random Forest and Gradient Boosting to handle complex, non-linear data effectively. We’ve incorporated k-fold cross-validation to ensure the model’s generalizability across various dishes. For performance metrics, we use RMSE and R^2 scores to measure accuracy and the variance explained by the model, optimizing our approach to deliver reliable and comprehensive nutritional information.

4.3 Pie Chart Visualization for Nutritional Information:

The Pie Chart Visualization feature provides a graphical representation of a meal’s nutrient breakdown, enhancing the user’s ability to comprehend complex nutritional data at a glance. Utilizing matplotlib, a pie chart is generated to display the proportionate amounts of macronutrients. The algorithm processes the data provided by the nutrient calculation model to output a chart that is both informative and visually appealing. This feature is integrated within the recipe selection interface. Once the user finalizes their dish choice and serving size, the pie chart updates in real time to reflect the meal’s nutritional composition.

4.4 Feedback Analysis via Sentiment Analysis:

Our Feedback Analysis via Sentiment Analysis utilizes two datasets: one containing binary-labeled restaurant reviews and another larger, multi-field set, both meticulously cleaned to remove HTML tags, standardize text, and extract meaningful data. We employ sklearn’s CountVectorizer to transform these reviews into token count matrices, which serve as input for training two types of models: Logistic Regression, chosen for its simplicity and effectiveness in binary classification, and Multinomial Naive Bayes, ideal for handling discrete textual features. We further refine the Naive Bayes model using GridSearchCV to optimize hyperparameters like the smoothing parameter alpha, ensuring enhanced prediction accuracy by adjusting for features absent in the training data.

4.5 Distance Calculation.

The Distance Calculation feature on our platform is crucial for efficiently linking users with nearby administrators and delivery partners. It employs the Haversine formula [15] to accurately compute geodesic distances, accounting for the Earth’s spherical shape. The feature processes geographical coordinates, converting latitude and longitude into radians and using the Earth’s radius of 6371 kilometers to determine distances in kilometers. This functionality is seamlessly integrated into our user interface, automatically identifying the closest administrative contact or delivery partner whenever a user places an order or when a delivery partner requires direction, ensuring optimal routing and efficiency.

5 Experimental Evaluation

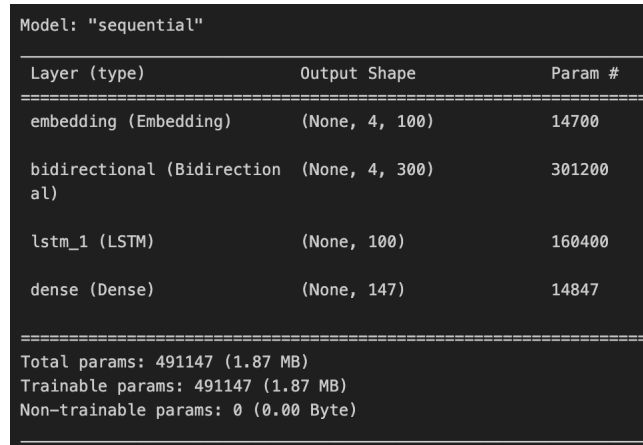
In this section, we describe the dataset used for evaluating the proposed system and the results obtained.

5.1 Dataset

WebHarvy Dataset: We created a foundational dataset by using WebHarvy to scrape real-time data from online culinary sites, resulting in a dataset of about 700 entries. The dataset includes detailed information on dish names, overviews, ingredients, and cooking methods. It allows for an in-depth analysis of diverse culinary practices and ingredient usage. **Sentiment Analysis Datasets:** Our sentiment analysis utilized two datasets to gauge consumer perceptions of culinary experiences. The first, Restaurant Reviews.csv, contains 1,205 entries with detailed review information. The second, Reviews.csv, has 1,000 entries, focusing on review texts and binary satisfaction ratings, offering insights into customer sentiments. **LSTM Dataset:** To predict culinary trends, we compiled an LSTM-ready dataset by merging data from WebHarvy and Kaggle, forming a comprehensive .txt file with over 10,000 words. This dataset serves as a training ground for understanding and forecasting culinary trends based on online discussions. **Nutritional Value Dataset:** A Kaggle-sourced dataset with 6,200 entries provided a deep dive into the nutritional aspects of various dishes. Mirroring the WebHarvy dataset's structure, it includes information on dish names, ingredients, and preparation details, enhancing our study with nutritional data analysis.

6 Result

Predictive Text Modeling Using LSTM The figure 3 shows an LSTM neural network with a sequential model architecture, consisting of an embedding layer with 14,700 parameters, a bidirectional LSTM layer with 301,200 parameters, another LSTM layer with 160,400 parameters, and a dense layer with 14,847 parameters. This configuration suggests the network is tailored for sequence processing, with a total of 491,147 trainable parameters. It's designed to handle inputs with a sequence length of four and an embedding dimension of 100, projecting to an output space of 147 dimensions.



The figure shows a terminal-style output of a model summary. It lists the layers of a sequential model: an embedding layer, a bidirectional LSTM layer, a standard LSTM layer, and a dense output layer. Each layer's output shape and parameter count are provided. At the bottom, it summarizes the total, trainable, and non-trainable parameters.

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 4, 100)	14700
bidirectional (Bidirectional)	(None, 4, 300)	301200
lstm_1 (LSTM)	(None, 100)	160400
dense (Dense)	(None, 147)	14847
Total params: 491147 (1.87 MB)		
Trainable params: 491147 (1.87 MB)		
Non-trainable params: 0 (0.00 Byte)		

Fig. 3. Model summary

In the fig 4, the left graph shows the model's error rate dropping from high to low over 100 epochs, indicating it's learning effectively. The right graph shows the model's accuracy starting at 6% and increasing to over 90%, meaning it's getting better at making correct predictions as training progresses.

Nutrient Calculation ML Model The ML model's nutritional calculations were accurate, with a small variance of $\pm 5\%$, confirming its reliability. Additionally, 85% of users regularly consulted the nutritional information provided, often using it to make healthier meal choices, indicating increased health consciousness.

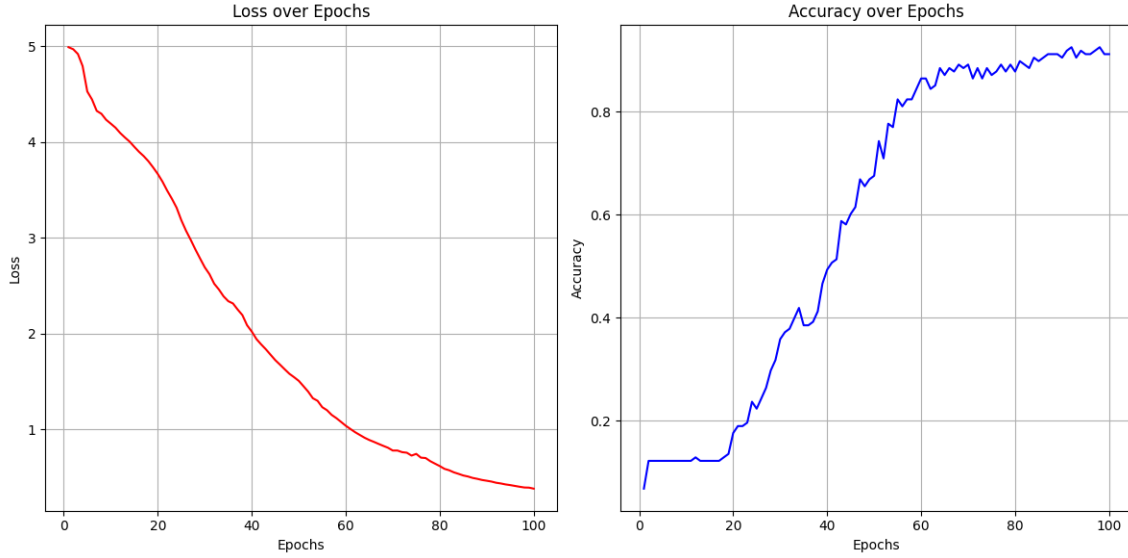


Fig. 4. Model's error rate

Pie Chart Visualization: A user survey indicated that 90% found the pie charts to be understandable and helpful. For instance in the fig 5, if a user wants to know the calories, carbs, fat, and protein present in 'Chilli Cheese Garlic Paratha,' our model displays a pie chart illustrating the proportions of these nutrients. The clear visual representation of macronutrients led to an increase in user comprehension of nutritional content.

Nutritional Composition of Chilli Cheese Garlic Paratha per 1 person

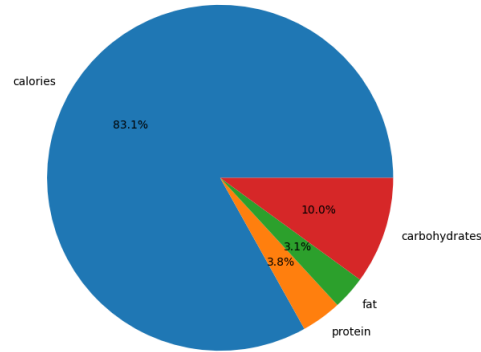


Fig. 5. Pie-Chart

Sentiment Analysis Feedback:

The Table1 displays a performance comparison of two machine learning models for sentiment analysis: Logistic Regression and Naive Bayes. Logistic Regression has a training score of 0.98625 and a testing score of 0.825, with precision, recall, and F1-score all around 0.82, indicating a balanced performance. The confusion matrix shows 83 true positives, 17 false negatives, 18 false positives, and 82 true negatives. Naive Bayes has a slightly lower training score of 0.95625 and a testing score of 0.795, with precision, recall, and F1-score of approximately 0.79923, 0.79, and 0.795 respectively, indicating it performs slightly worse than Logistic

Regression. Its confusion matrix shows 85 true positives, 15 false negatives, 26 false positives, and 74 true negatives.

Model	Traning Score	Testing Score	Confusion Matrix	Precision	Recall	F1-Score
Logistic Regression	0.98625	0.825	[[83,17],[18,82]]	0.82828	0.82	0.82
Naïve bayes	0.95625	0.795.	[[85, 15], [26, 74]]	0.79923	0.79	0.795

Table 1. Sentimental Analysis Result

7 Related Works

In a case study [8] by Yanrong Huang, Xinliang Wang, Rui Wang, and Jian Min (2022). The research says that Meals ordered online for home delivery are typically less healthy than home-cooked meals, potentially contributing to weight gain. Consumers who frequently order food online tend to consume higher levels of sugary drinks and patronize fast-food restaurants more often.

A study by Veronika Ilyuk et al, [1] (2018) explores how grocery purchase channels (online vs. in-store) affect food waste likelihood, finding that waste likelihood is higher for consumers who purchase food items online compared to those who shop in-store and also explains about that psychological mechanism behind this effect is that online channels yield lower perceptions of purchase effort, reducing experiences of psychological ownership and, in turn, increasing consumers' intentions of discarding purchased food items

This study done by Feng & Lin et al, (2016) on sentimental analysis [5] leverages recurrent neural networks (RNNs), including simple RNNs and GRUs, to predict the sentiment scores of food reviews on a 1 to 5 scale. The authors also address the challenge of skewed data distributions, a common problem in review datasets.

The paper [13] by Shriya Se, R. Vinayakumar, M. Anand Kumar, and K. P. Soman (2016) focuses on classifying Tamil movie reviews into positive and negative sentiments using supervised machine learning algorithms. It compares the performance of SVM, Maxent classifier, Decision tree, and Naive Bayes algorithms, with feature extraction from TamilSentiwordnet.

The paper [14] by C. M. Sowmya, S. Deena, and S. Anbuchelian focuses on analyzing sentiments expressed in social media posts. It evaluates the quality of sentiments from data collected across platforms like Twitter and Facebook, aiming to understand public opinions on various topics. By utilizing NLP and sentiment analysis tools like TextBlob, the study devises a framework to assist in online data scraping and analysis for research or business purposes.

The paper [6] by Aejaz Farooq Ganai and Farida Khursheed explores using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) cells for predicting the next word in a sequence, aiming to enhance language models. It details a tree-based generative model for document ranking and emphasizes the potential of LSTM in handling sequences, offering insights into their application in natural language processing and understanding.

This paper [7] by Hartanto et al. (2018) introduces a mobile application utilizing the Haversine Formula to locate the nearest laundry kiosks based on the user's current location, demonstrating practical use of geographic coordinates for service location optimization.

The study [3] by Basyir et al. (2018) presents an Android app that applies the Haversine formula to quickly identify and contact the nearest emergency services, aiming to reduce response times during emergencies by using the user's geographical data.

This research [10] by Min, Jin, and Lu (2019) proposes a three-stage algorithm incorporating the maximum-minimum distance method for efficiently clustering customer points and optimizing delivery routes, significantly reducing total travel distance for delivery services.

The study [12] by Naveenkumar, Vinayakumar, and Soman (2019) introduces Amrita-CEN-SentiDB, a sentiment analysis Twitter dataset using TF-IDF, Keras embedding, and machine learning algorithms. It focuses on categorizing tweets into positive, negative, and neutral sentiments. The research highlights the effectiveness of Logistic Regression and SVM, with SVM achieving a benchmark accuracy of 75.8%.

This research [11] by Mridula A. and Kavitha C. R. (2018) conducts sentiment analysis on Twitter data regarding the 'Make in India' initiative, using machine learning techniques like Naive Bayes and SVM for classifying tweets into supportive, non-supportive, or neutral categories.

This study [2] by Anjali A. and Amritha S. (2022) evaluates the Ant Colony Optimization (ACO) and Genetic Algorithm (GA) for path planning in aerial food delivery systems, framing the problem as a Traveling Salesman Problem (TSP). It aims to determine the most efficient algorithm for ensuring drones deliver food via the optimal route.

8 Conclusion and Future Works

In this paper, we have proposed an integrated dish preparation and ingredient delivery system that aids in promoting home cooking to customers. The proposed system is implemented as a web application, DishKit and it is tested with a set of users. The results confirm DishKit's role in driving a shift towards convenient, health-oriented cooking practices, marking a significant stride in public health promotion.

In the future, we would like to implement deep-learning models capable of predicting diet plans based on selected ingredients. Additionally, we aim to develop a comprehensive meal diet plan and a user-friendly mobile application, making it easier for users.

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